

1. Administrative & Study Identification

Study Title: A Study to Assess the Long-term Safety of KarXT for the Treatment of Manic Episodes in Bipolar-I Disorder (BALSAM-3)
Official Full Title: A Phase 3, Open-label Extension Study to Assess the Long-term Safety of KarXT for the Treatment of Mania or Mania With Mixed Features in Bipolar-I Disorder (BALSAM-3)
Short Title/Acronym: BALSAM-3 **Protocol Number:** CN012-0038 **NCT Number:** NCT06929273 **Posting ID:** 39271850582697095 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** KarXT (Xanomeline and trospium chloride) **Phase of Development:** Phase 3 **Trial Status:** RECRUITING **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and tolerability of KarXT in the treatment of participants with mania or mania with mixed features associated with Bipolar-I disorder (BP-I). **Secondary Objectives:** To assess the long-term safety profiles, specifically focusing on adverse events of special interest (AESIs), serious adverse events (SAEs), and TEAEs leading to treatment discontinuation in participants with bipolar mania over an extended duration.

2.2 Background & Rationale To address the need for effective, long-term treatments for manic episodes in BP-I that minimize the side effects typical of standard atypical antipsychotics. KarXT utilizes a novel mechanism combining a muscarinic agonist (xanomeline) with a peripheral muscarinic antagonist (trospium) to treat mania symptoms without relying on traditional dopaminergic pathways.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Open-label extension) **Blinding:** Open-label **Randomization:** N/A (All participants receive KarXT)

3.2 Planned Enrollment & Duration Targeted Enrollment: 450 participants (derived from rollover participants from parent trials and newly enrolled de novo participants). **Number of Sites:** Approximately 69 global locations **Estimated Study Start:** Mid-2024 **Estimated Primary Completion:** March 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. [Rollover Participants] Participants must have completed the treatment period of a parent double-blind placebo-controlled study (CN0120036, CN0120037, or CN0120046). 2. [De Novo Participants] Primary diagnosis of Bipolar-I disorder established by a comprehensive psychiatric evaluation based on DSM-5-TR criteria and confirmed by the Mini International Neuropsychiatric Interview (MINI, v7.0.2), with symptoms of mania or mixed mania. 3. [De Novo Participants] Young Mania Rating Scale (YMRS) score of ≥ 14 at Screening and at baseline. 4. [De Novo Participants] CGI-BP score of ≥ 3 at Screening and at baseline. 5. [De Novo Participants] Does not require hospitalization for acute mania.

4.2 Exclusion Criteria 1. [Rollover Participants] Discontinuation from any KarXT parent studies (CN0120036, CN0120037, or CN0120046). 2. [De Novo Participants] Risk for suicidal behavior at baseline as determined by Investigator's clinical assessment or history of suicidal behavior as assessed on C-SSRS. 3. [De Novo Participants] Primary diagnosis of BP-I with rapid cycling (≥ 4 distinct mood episodes in one year). 4. [De Novo Participants] Any primary DSM-5-TR disorder other than BP-I with mania or mania with mixed features within 12 months before Screening (confirmed using MINI version 7.0.2 at Screening), including BP-I with depression (previous 3 months only), Bipolar-II disorder, major depressive disorder, borderline personality disorder, and primary psychotic disorder, with the exception of mild anxiety disorders. 5. [De Novo Participants] Diagnosis of moderate to severe substance use disorder (except tobacco use disorder) within 12 months before Screening, or current use as determined by urine toxicology screen or alcohol test. 6. [De Novo Participants] History of irritable bowel syndrome (with or without constipation), serious constipation requiring treatment within the last 6 months. 7. [De Novo Participants] History or high risk of urinary retention, gastric retention, or untreated narrow-angle glaucoma. 8. Other protocol-defined Inclusion/Exclusion criteria apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: KarXT continuous oral administration (dosed according to protocol titration and maintenance parameters). **Route of Administration:** Oral (Capsule) **Duration of Treatment:** Long-term extension phase.

5.2 Concomitant & Prohibited Therapy Permitted: Protocol-allowed standard of care (e.g., specific mood stabilizers depending on adjunctive protocol allowances). **Prohibited:** Concurrent investigational psychiatric therapies, specific concomitant medications with strong anticholinergic properties that could exacerbate the mechanism of action.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent,
Psychiatric Eval (MINI, YMRS, CGI-BP, C-SSRS), Safety Labs | |
Baseline | Day 0 | Enrollment (De Novo) or Roll-over
Confirmation, First Dose, Vital Signs | | **Interim** | Periodic |
Safety Labs, Efficacy Assessments (YMRS, CGI-BP), AE tracking | |
End of Study | Final Assessment | Final Vital Signs, Exit
Interview, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes * Number of participants with treatment emergent adverse events (TEAEs).

7.2 Secondary Outcomes * Number of participants with AEs of special interest (AESIs). * Number of participants with serious AEs (SAEs). * Number of participants with TEAEs leading to treatment discontinuation.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient interview, psychiatric check-ins, and safety protocols at designated visits. **Serious Adverse Events (SAEs):** Required reporting timeline, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External safety committee oversight to evaluate long-term cumulative safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who received at least one dose of KarXT in the open-label phase). **Sample Size Justification:** Sufficient power (targeted enrollment of 450) scaled to characterize the long-term safety profile and document rare AEs in BP-I populations rolling over from initial trials and de novo enrollees. **Handling of Missing Data:** Mixed-effects Model for Repeated Measures (MMRM) or Multiple Imputation applied for longitudinal efficacy measurements where appropriate.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote clinical monitoring visits to review compliance. **Audit Trail:** Complete eCRF tracking,

psychiatric scale rater verification, and source data verification (SDV).

11. Study Identifiers & Governance

Primary ID: CN012-0038 **Registry Number:** NCT06929273 **Posting ID:** 39271850582697095 **Company / Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Assess the Long-term Safety of KarXT for the Treatment of Manic Episodes in Bipolar-I Disorder (BALSAM-3) **Official Regulatory Title:** A Phase 3, Open-label Extension Study to Assess the Long-term Safety of KarXT for the Treatment of Mania or Mania With Mixed Features in Bipolar-I Disorder (BALSAM-3)

12. Clinical Framework (Bipolar-I Specific)

Disease Areas: Bipolar Disorder Type I With Mania **Preferred UMLS Name:** Bipolar I Disorder, Manic **Primary Condition:** Bipolar-I Disorder with Manic Episodes **Diagnosis Threshold:** DSM-5-TR criteria, YMRS score ≥ 14 , CGI-BP score ≥ 3 **Medical Methodology:** Muscarinic Receptor Targeted Therapy **Optimization Target:** Maintenance of symptom improvement and KarXT tolerability

13. Study Procedures & Methodology

Management Pathway: Long-term psychiatric monitoring and management **Education Protocol (HCP):** Psychiatric scale calibration and KarXT specific safety monitoring training **Education Protocol (Patient):** Adherence monitoring and side effect tracking **Quality Audit Mechanism:** Rater reliability verification & patient compliance recording

14. Observation & Follow-Up Schedule

Total Duration: Long-term extension evaluation **Onsite Visit Cadence:** Regular maintenance protocol visits **Remote Monitoring Frequency:** As required per clinical assessment schedule **Checklist Review Interval:** Standard interval psychiatric scale assessments

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	06-Apr-2026				
Clinical Operations Lead	[Name]	_____	06-Apr-2026		Lead		
Statistician	[Name]	_____	06-Apr-2026				